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Introduction to Programming (Practical)

Provided by **VU Answer**

Q1) Define constructor

Answer:

A constructor is a special function in a class that is called when an object of that class is created. It is used to initialize the object's members.

Q2) Define ptr

Answer:

ptr is short for pointer, which is a variable that stores the memory address of another variable.

Q3) Arithmetic operator program

Answer:

```
#include <iostream>
using namespace std;
int main() {
    int a = 10, b = 5;
    cout << "a + b = " << a + b << endl;
    cout << "a - b = " << a - b << endl;
    cout << "a * b = " << a * b << endl;
    cout << "a / b = " << a / b << endl;
    cout << "a % b = " << a % b << endl;
    return 0;
}
```

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Q4) Write Array simple program

Answer:

```
#include <iostream>
using namespace std;
int main() {
    int arr[5] = {1, 2, 3, 4, 5};
    for (int i = 0; i < 5; i++) {
        cout << "Element " << i << ": " << arr[i] << endl;
    }
    return 0;
}
```

Q5) Write matrix simple program

Answer:

```
#include <iostream>
using namespace std;
int main() {
    int matrix[2][2] = {{1, 2}, {3, 4}};
    for (int i = 0; i < 2; i++) {
        for (int j = 0; j < 2; j++) {
            cout << matrix[i][j] << " ";
        }
        cout << endl;
    }
    return 0;
}
```

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```
}
```

Q6) Write inline function *max* simple program

Answer:

```
#include <iostream>
using namespace std;
inline int max(int a, int b) {
    return (a > b) ? a : b;
}
int main() {
    int x = 10, y = 20;
    cout << "Max value is: " << max(x, y) << endl;
    return 0;
}
```

Q7) Write Recursive function program.

Answer:

```
#include <iostream>
using namespace std;
int factorial(int n) {
    if (n == 0)
        return 1;
    else
        return n * factorial(n - 1);
}
int main() {
```

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```
int num = 5;
cout << "Factorial of " << num << " is " << factorial(num) << endl;
return 0;
}
```

Q8) Create Class with volume name and also define their members

Answer:

```
class Volume {
private:
    int length;
    int width;
    int height;
public:
    Volume(int l, int w, int h) : length(l), width(w), height(h) {}
    int calculateVolume() {
        return length * width * height;
    }
};
```

Q9) Create area and circumference calculate

Answer:

```
#include <iostream>
using namespace std;
class Circle {
private:
    double radius;
```

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public:

```
Circle(double r) : radius(r) {}  
double area() {  
    return 3.14 * radius * radius;  
}  
double circumference() {  
    return 2 * 3.14 * radius;  
}
```

};

```
int main() {  
    Circle c(5.0);  
    cout << "Area: " << c.area() << endl;  
    cout << "Circumference: " << c.circumference() << endl;  
    return 0;  
}
```

Q10) Write program for calloc & malloc function

Answer:

```
#include <iostream>  
using namespace std;  
int main() {  
    int *ptr;  
    int n = 5;  
    // Using malloc  
    ptr = (int*)malloc(n * sizeof(int));  
    for (int i = 0; i < n; i++) {
```

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```
ptr[i] = i + 1;
cout << ptr[i] << " ";
}
free(ptr);
cout << endl;

// Using calloc
ptr = (int*)calloc(n, sizeof(int));
for (int i = 0; i < n; i++) {
    ptr[i] = i + 1;
    cout << ptr[i] << " ";
}
free(ptr);
return 0;
}
```

Q11) Write a program to find area of circle and circumference

Answer:

```
#include <iostream>
#define PI 3.14159

int main() {
    float radius, area, circumference;
    std::cout << "Enter the radius of the circle: ";
    std::cin >> radius;
    area = PI * radius * radius;
    circumference = 2 * PI * radius;
```

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```
std::cout << "Area of the circle: " << area << std::endl;
std::cout << "Circumference of the circle: " << circumference << std::endl;
return 0;
}
```

Q12) Ik program mn array Ki values de hwe the or us values ko copy KR KR new array bnana the or Jo new array se values ae usko display krwana Tha

Answer:

```
#include <iostream>
int main() {
    const int SIZE = 5;
    int originalArray[SIZE] = {10, 20, 30, 40, 50};
    int newArray[SIZE];
    for (int i = 0; i < SIZE; i++) {
        newArray[i] = originalArray[i];
    }
    std::cout << "Values of the new array are: ";
    for (int i = 0; i < SIZE; i++) {
        std::cout << newArray[i] << " ";
    }
    return 0;
}
```

OUTPUT

Values of the new array are: 10 20 30 40 50

Q13) Write a program which perform product of first 10 numbers through recursion.

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Answer:

```
#include <iostream>

int product(int n) {
    if (n == 1)
        return 1;
    else
        return n * product(n - 1);
}

int main() {
    int result = product(10);
    std::cout << "Product of the first 10 numbers: " << result << std::endl;
    return 0;
}
```

Q14) Array simple program

Answer:

```
#include <iostream>

int main() {
    const int SIZE = 5;
    int array[SIZE] = {1, 2, 3, 4, 5}; // Initialize array
    std::cout << "Array elements are: ";
    for (int i = 0; i < SIZE; i++) {
        std::cout << array[i] << " "; // Print each element
    }
    return 0;
}
```


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Q15) Write matrix simple program

Answer:

```
#include <iostream>

int main() {
    const int rows = 2, columns = 2;
    int matrix[rows][columns] = {{1, 2}, {3, 4}}; // Initialize 2x2 matrix
    std::cout << "Matrix elements are: " << std::endl;
    for (int i = 0; i < rows; i++) {
        for (int j = 0; j < columns; j++) {
            std::cout << matrix[i][j] << " ";
        }
        std::cout << std::endl;
    }
    return 0;
}
```

Q16) inline function *max* program

Answer:

```
#include <iostream>

// Inline function to find maximum of two numbers
inline int max(int a, int b) {
    return (a > b) ? a : b;
}

int main() {
    int num1 = 10, num2 = 20;
```

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```
std::cout << "Maximum number is: " << max(num1, num2) << std::endl;
return 0;
}
```

Q17) Recursive function program tha.

Answer:

```
#include <iostream>
int factorial(int n) {
    if (n <= 1)
        return 1;
    else
        return n * factorial(n - 1);
}
int main() {
    int num = 5;
    std::cout << "Factorial of " << num << " is: " << factorial(num) << std::endl;
    return 0;
}
```

Q18) What happens when we use new and delete operator?

Answer:

- The new operator allocates memory on the heap and returns a pointer to it.
- The delete operator deallocates memory that was previously allocated with new.
- If you forget to use delete, it results in a memory leak, meaning that the memory remains allocated even after the program ends.

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Q19) Write a program that print the sum of 1 to 1000 odd numbers using while loop.

Answer:

```
#include <iostream>

int main() {
    int sum = 0, number = 1;
    while (number <= 1000) {
        if (number % 2 != 0) { // Check if the number is odd
            sum += number;    // Add the odd number to sum
        }
        number++;
    }
    std::cout << "Sum of odd numbers from 1 to 1000 is: " << sum << std::endl;
    return 0;
}
```

Q20) Create call by reference program

Answer:

```
#include <iostream>
using namespace std;

void swap(int &a, int &b) {
    int temp = a;
    a = b;
    b = temp;
}

int main() {
    int x = 5, y = 10;
```

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```
cout << "Before swap: x = " << x << ", y = " << y << endl;
swap(x, y);
cout << "After swap: x = " << x << ", y = " << y << endl;
return 0;
}
```

Q21) Create setfill or set base template program and write output

Answer:

```
#include <iostream>
#include <iomanip>
using namespace std;
int main() {
    int num = 100;
    cout << setfill('*') << setw(10) << num << endl;
    cout << setbase(16) << num << endl;
    return 0;
}
```

Q22) Write output from this code

```
#include<iostream>
```

```
Using namespace std;
```

```
Int num1= 34, int num2 = 45 int= num3 = 46
```

```
If(num1 >num2&& num 1>num3)
```

```
Cout<<number 1 is greater ;
```

```
Else if (num 2 >num1 && num2>num3
```

```
Cout<< number 2 is greater;
```

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Else if (num3>num1 & num3 >num2)

Cout<<number 3 is greater;

Else

Cout<<All number are equal;

Answer:

Number 3 is greater

Q23) Write output in this code

Int num1=4,num2=5,fact 1= 1 fact2= 1

While (num1>1)

{

Fact1= fact 1*num1;

Num1 = num 1-1;

}

Cout<< the function of number 1 is <<fact 1<<endl;

While

Fact2= fact2*num2;

Num1 = num 2-1;

}

Cout<< the function of number 2 is <<fact 2<<endl;

If(fact >fact2)

Cout<<the factorial of number 1 is greater than 2

Cout<<the factorial of number 2 is greater than 1

Answer:

The factorial of number 1 is 24

The factorial of number 2 is 120

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The factorial of number 2 is greater than 1

Q24) Write output

Template <class T>

Class first {

Public:

First ()

Cout<<created/n:

}

First ()

Cout<<destroyed/n:

}

Template <class u>

Class second {

Public:

Second{Cout<<created/n:

}

Second ()

Cout<<destroyed/n:

}

Answer:

Created

Created

Destroyed

Destroyed

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Q25) Write max 2 int program.

Answer:

```
#include <iostream>
using namespace std;
int max(int a, int b) {
    return (a > b) ? a : b;
}
int main() {
    int x = 10, y = 20;
    cout << "Max value is: " << max(x, y) << endl;
    return 0;
}
```

Q26) Overload function square for integer and float data type. The overload function will take a single arguments square.

Answer:

```
#include <iostream>
using namespace std;
int square(int x) {
    return x * x;
}
float square(float x) {
    return x * x;
}
int main() {
    int intVal = 5;
    float floatVal = 5.5;
```

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```
cout << "Square of " << intVal << " is " << square(intVal) << endl;
cout << "Square of " << floatVal << " is " << square(floatVal) << endl;
return 0;
}
```

Q27) Write program find sum odd number 1 to 1000 by using while loop.

Answer:

```
#include <iostream>
using namespace std;
int main() {
    int sum = 0;
    int i = 1;
    while (i <= 1000) {
        if (i % 2 != 0) {
            sum += i;
        }
        i++;
    }
    cout << "Sum of odd numbers from 1 to 1000 is " << sum << endl;
    return 0;
}
```

Q28) Write program print matrix shape element[2][2] by using class Matric.

Answer:

```
#include <iostream>
using namespace std;
```


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```
class Matrix {  
    private:  
        int mat[2][2];  
    public:  
        void setMatrix(int a[2][2]) {  
            for (int i = 0; i < 2; i++) {  
                for (int j = 0; j < 2; j++) {  
                    mat[i][j] = a[i][j];  
                }  
            }  
        }  
        void printMatrix() {  
            for (int i = 0; i < 2; i++) {  
                for (int j = 0; j < 2; j++) {  
                    cout << mat[i][j] << " ";  
                }  
                cout << endl;  
            }  
        }  
};  
  
int main() {  
    Matrix m;  
    int a[2][2] = {{1, 2}, {3, 4}};  
    m.setMatrix(a);  
    m.printMatrix();  
    return 0;  
}
```

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```
}
```

Q29) Two-dimensional array program

Answer:

```
#include <iostream>
using namespace std;
int main() {
    int matrix[3][3] = {
        {1, 2, 3},
        {4, 5, 6},
        {7, 8, 9}
    };
    for (int i = 0; i < 3; i++) {
        for (int j = 0; j < 3; j++) {
            cout << matrix[i][j] << " ";
        }
        cout << endl;
    }
    return 0;
}
```

```
Q30) #include <iostream>
using namespace std;
template<class T,int max>
int arrayMax(T arr[], int n)
{
```

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```
int m = max;
for(int i=0 ; i<n ; i++)
if(arr[i]>m)
m= arr[i];
return m;
}
int main()
{
int array1[]={10,20,15,12};
int n1 = sizeof(array1)/sizeof(array1[0]);
char array2[]={1,2,3};
int n2 = sizeof(array2)/sizeof (array2[0]);
cout << arrayMax < int,10000>(array1,n1)<< endl;
cout << arrayMax < char,256>(array2,n2);
return 0;
}
```

Write Output

Answer:

10000

256

Q31) Variable in C++.

Answer:

A variable in C++ is a storage location with a specific type that determines what kind of data it can hold.

Example: int x = 5;

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Q32) Write a program that print the sum of 1 to 1000 odd numbers using while loop

Answer:

```
#include <iostream>
using namespace std;
int main() {
    int sum = 0;
    int i = 1;
    while (i <= 1000) {
        if (i % 2 != 0) {
            sum += i;
        }
        i++;
    }
    cout << "Sum of odd numbers from 1 to 1000 is " << sum << endl;
    return 0;
}
```

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